

Limitations of K-means Clustering →

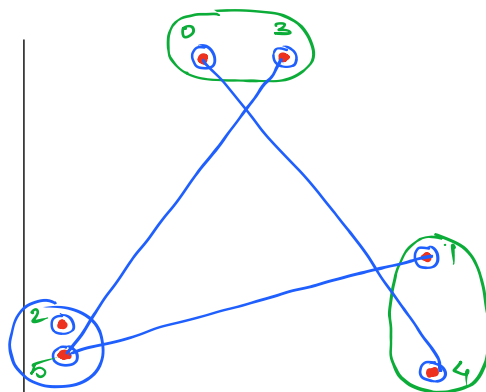
- ① Pre-defined k-value.
- ② Affected by the outliers.
- ③ It can only identify spherical/circular clusters because of centroid values.
- ④ Sometimes the elbow point graph does not give us the correct idea about the k-value.

Hierarchical Clustering →

- ① Agglomerative clustering approach. ✓
- ② Divisive clustering approach.

Agglomerative approach

X = 6 datapoints
K = ?



Linkage
Single
→ Complete
Average
ward (SSD)

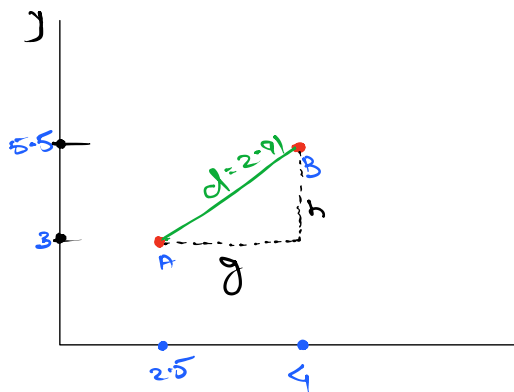
Proximity matrix

	0	1	2	3	4	5	6
0		0					
1							
2							
3							
4							
5							
6							

For distance calculation we have 3 different methods available:

- ① Euclidean distance.
- ② Manhattan distance.
- ③ Minkowski distance.

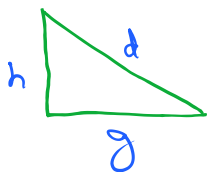
Euclidean dist



$$\begin{aligned} A &= (x_1, y_1) \\ &= (2.5, 3) \\ B &= (x_2, y_2) \\ &= (4, 5.5) \end{aligned}$$

$$\begin{aligned} g &= 4 - 2.5 \\ &= 1.5 \end{aligned}$$

$$\begin{aligned} h &= 5.5 - 3 \\ &= 2.5 \end{aligned}$$



$$\begin{aligned} d^2 &= h^2 + g^2 \\ d &= \sqrt{h^2 + g^2} \end{aligned}$$

$$\begin{aligned} d &= \sqrt{g^2 + h^2} \\ &= \sqrt{(1.5)^2 + (2.5)^2} \\ &= 2.91 \end{aligned}$$

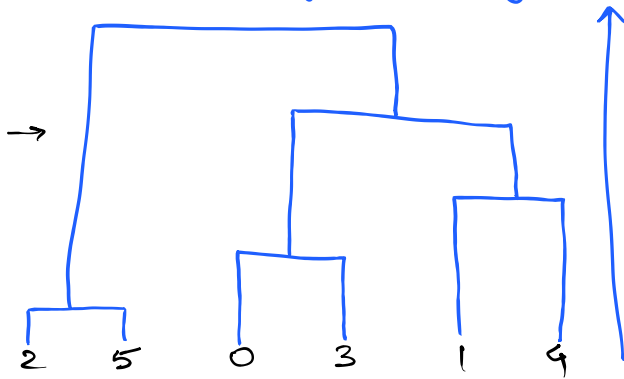
↑
Euclidean
Distance

Manhattan Dist →

$$\begin{aligned} d &= |g| + |h| \\ &= 1.5 + 2.5 \\ &= 4 \end{aligned}$$

Dendrogram to represent the step-wise clusterings as a graph →

Find the number of clusters →



Types of Linkage functions →

- ① Single Linkage.
- ② Complete Linkage.
- ③ Average Linkage.
- ④ Word Linkage.

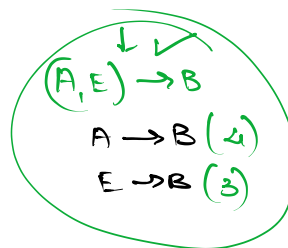
Mathematical working of Agglomerative Clustering

	A	B	C	D	E
A	0	4	7	9	①
B	4	0	3	5	3
C	7	3	0	2	6
D	9	5	2	0	8
E	1	3	6	8	0

This is our initial proximity matrix b/w 5 points A, B, C, D, E as an example.

→

	(A, E)	B	C	D
(A, E)	0	4	7	9
B	4	0	3	5
C	7	3	0	2
D	9	5	2	0



$$(A, E) \rightarrow (C, D)$$

$$(A, E) \rightarrow C$$

$$(A, E) \rightarrow D$$

	$(A, E) \rightarrow B$	(C, D)
(A, E)	0	4
B	4	0
(C, D)	9	5

$$((A, E), B) \rightarrow (C, D)$$

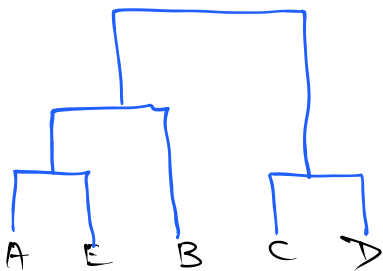
$$(A, E) \rightarrow (C, D)$$

$$B \rightarrow (C, D)$$

	$((A, E), B)$	(C, D)
$((A, E), B)$	0	9
(C, D)	9	0

$$[[A, E], B], [C, D]$$

↓



→

Find the
K-value.

